

41.7 Summary and Notes

41.7.1 Summary

Idea.

1. The pattern of missingness is data. Writing the joint law as a model for the complete array times a mechanism Eq. 41.1.1 separates what the analyst wants from how the holes arose, and sorts mechanisms into MCAR, MAR and MNAR (Definition 41.1.1).
2. Nothing in the observed data distinguishes MAR from MNAR: every choice of the conditional distribution of the unrecorded values given the recorded ones gives the same observed-data likelihood (Proposition 41.1.1). MCAR against MAR is testable; MAR against MNAR is not.
3. A complete-case analysis estimates functionals of the law given completeness. It is valid under MCAR, and — the useful case — valid for a correctly specified regression whenever the missingness depends on the covariates but not on the response (Proposition 41.2.1). It is not generally valid under MAR.
4. Mean imputation deflates the variance by the fraction recorded (Proposition 41.2.2); regression imputation without noise inflates correlations; any single imputation understates every standard error; available-case correlation matrices need not be nonnegative definite.
5. Under MAR with distinct parameters the mechanism is ignorable: likelihood, likelihood ratio tests and Bayesian posteriors may be computed from the observed-data likelihood alone (Theorem 41.3.1). The licence costs a model for any variable that can go missing, including covariates. For a monotone pattern that likelihood factors into conditional models on nested subsamples (Exercise (B1) of [Section 41.1](#) and Exercise (B1) of [Section 41.3](#)).
6. EM maximizes the observed-data likelihood by alternating a conditional expectation of the complete-data log-likelihood with a complete-data fit (Definition 41.4.1). Every sweep increases the observed-data likelihood (Theorem 41.4.1), and fixed points are stationary points, not necessarily maxima.
7. For missing responses in a linear model, EM returns the observed-data least squares estimate and $\text{SSE}_o/(n - m)$ (Proposition 41.4.2), deriving the degrees-of-freedom correction that Theorem 18.6.1(e) asserted. For a missing covariate the E step must condition on the response (Proposition 41.4.3).
8. Observed information equals expected complete-data information minus the variance of the complete-data score (Proposition 41.4.4). That difference sets both EM's rate of convergence and the amount by which the complete-data information understates the variance; standard errors come from Louis's formula, numerical differentiation or the bootstrap, never from the matrix the M step inverts.
9. Multiple imputation draws M completions from the posterior predictive distribution (Definition 41.5.1), analyses each, and combines by $\bar{Q}$ and $T = \bar{U} + (1 + M^{-1})B$ on $(M - 1)/\gamma^2$ degrees of freedom (Theorem 41.5.1). A finite M inflates the standard error by $(1 + \gamma_\infty/M)^{1/2}$. The imputation model must contain everything the analysis will use, above all the response.
10. Under MNAR the selection and pattern-mixture factorizations describe the same

unidentified object (Proposition 41.6.1). A delta-adjustment names the unidentified part, every value of it fits the observed data equally well, and the tipping point reports how far the assumption must move before the conclusion does.

41.7.2 Notes and sources

The framework. The classification into MCAR, MAR and MNAR, the selection factorization and the ignorability theorem are due to Rubin (1976); the book-length development is Little and Rubin (2019), whose organization this chapter follows, the proofs, examples and numbers here being the book's own. Schafer (1997) treats the multivariate normal and categorical cases, including the monotone factorizations of [Section 41.3](#), whose closed form is Anderson's (1957). The realized-value and everywhere versions of MAR are separated by Seaman, Galati, Jackson and Carlin (2013); the test of MCAR is Little (1988), and the general form of Proposition 41.1.1 is Molenberghs, Beunckens, Sotito and Kenward (2008).

Complete cases. Part (b) of Proposition 41.2.1 has been rediscovered many times; the clearest statements are Little (1992) and White and Carlin (2010), who also note that complete-case analysis can beat multiple imputation for efficiency when the mechanism depends only on co-variates. Inverse-probability weighting descends from Horvitz and Thompson (1952), the doubly robust augmentation from Robins, Rotnitzky and Zhao (1994).

EM. The algorithm was named and given its general form by Dempster, Laird and Rubin (1977), who also identified the rate of convergence as the largest fraction of missing information; special cases are older, the iteration of Theorem 18.6.1(c) being the missing-plot scheme of Healy and Westmacott (1956), built on Yates's (1933) formula. The missing-information principle is Orchard and Woodbury (1972), and Eq. 41.4.5 is Louis's (1982). Wu (1983) corrected the convergence claims of the original paper; Meng and Rubin (1991) extract the observed information from the EM iterations themselves, and McLachlan and Krishnan (2008) is the book-length treatment.

Multiple imputation. The idea is Rubin's (1978) and the definitive statement is Rubin (1987), where the combining rules, the degrees of freedom and the efficiency calculation of Theorem 41.5.1 all appear; the small-sample degrees of freedom are Barnard and Rubin (1999). Congeniality is Meng's (1994), whose discussion in the same issue remains the best entry point to when Rubin's variance estimator is consistent. Chained equations were developed by Van Buuren, Boshuizen and Knook (1999) and by Raghunathan, Lepkowski, Van Hoewyk and Solenberger (2001); van Buuren (2018) is the practical handbook.

MNAR and sensitivity. Selection models in the econometric tradition begin with Heckman (1979), pattern-mixture models with Little (1993), who introduced the complete-case missing-value restriction of Exercise (C1). The non-identifiability in Proposition 41.6.1(c) is why Manski's (1990) assumption-free bounds, the subject of Exercise (C2), remain worth computing. Delta-adjustment and tipping-point analyses became standard in clinical trials after the National Research Council (2010) report, which is short and unusually clear about the difference between preventing missing data and repairing it. The reference-based schemes named in [Section 41.6](#) are due to Carpenter, Roger and Kenward (2013).

Where this goes next. Nothing here is special to a normal response: ignorability, EM and Rubin's rules apply verbatim to the generalized linear models of Chapters 34 to 37, with harder integrals and an E step usually done by simulation. The estimating-equation methods of [Chapter 40](#) stay consistent when the missingness is completely at random or depends only

on modelled, fully observed covariates, and become inconsistent when it depends on previously observed responses — the MAR case in which likelihood remains valid — because the estimating function $E\{R_i(Y_i - \mu_i)\}$ is then no longer zero. The repair is inverse-probability weighting.

41.7.3 References

- Anderson, T. W. (1957). Maximum Likelihood Estimates for a Multivariate Normal Distribution When Some Observations Are Missing. *Journal of the American Statistical Association* 52(278), 200–203.
- Barnard, John and Rubin, Donald B. (1999). Small-Sample Degrees of Freedom with Multiple Imputation. *Biometrika* 86(4), 948–955.
- Carpenter, James R., Roger, James H. and Kenward, Michael G. (2013). Analysis of Longitudinal Trials with Protocol Deviation: A Framework for Relevant, Accessible Assumptions, and Inference via Multiple Imputation. *Journal of Biopharmaceutical Statistics* 23(6), 1352–1371.
- Dempster, Arthur P., Laird, Nan M. and Rubin, Donald B. (1977). Maximum Likelihood from Incomplete Data via the EM Algorithm. *Journal of the Royal Statistical Society, Series B* 39(1), 1–38.
- Healy, Michael and Westmacott, Michael (1956). Missing Values in Experiments Analysed on Automatic Computers. *Journal of the Royal Statistical Society, Series C (Applied Statistics)* 5(3), 203–206.
- Heckman, James J. (1979). Sample Selection Bias as a Specification Error. *Econometrica* 47(1), 153–161.
- Horvitz, D. G. and Thompson, D. J. (1952). A Generalization of Sampling Without Replacement from a Finite Universe. *Journal of the American Statistical Association* 47(260), 663–685.
- Little, Roderick J. A. (1988). A Test of Missing Completely at Random for Multivariate Data with Missing Values. *Journal of the American Statistical Association* 83(404), 1198–1202.
- Little, Roderick J. A. (1992). Regression with Missing X's: A Review. *Journal of the American Statistical Association* 87(420), 1227–1237.
- Little, Roderick J. A. (1993). Pattern-Mixture Models for Multivariate Incomplete Data. *Journal of the American Statistical Association* 88(421), 125–134.
- Little, Roderick J. A. and Rubin, Donald B. (2019). *Statistical Analysis with Missing Data*. 3rd edition. Hoboken, NJ: Wiley.
- Louis, Thomas A. (1982). Finding the Observed Information Matrix When Using the EM Algorithm. *Journal of the Royal Statistical Society, Series B* 44(2), 226–233.
- Manski, Charles F. (1990). Nonparametric Bounds on Treatment Effects. *American Economic Review* 80(2), 319–323.
- McLachlan, Geoffrey J. and Krishnan, Thiriyambakam (2008). *The EM Algorithm and Extensions*. 2nd edition. Hoboken, NJ: Wiley.
- Meng, Xiao-Li (1994). Multiple-Imputation Inferences with Uncongenial Sources of Input. *Statistical Science* 9(4), 538–558.
- Meng, Xiao-Li and Rubin, Donald B. (1991). Using EM to Obtain Asymptotic Variance–Covariance Matrices: The SEM Algorithm. *Journal of the American Statistical Association* 86(416), 899–909.
- Molenberghs, Geert, Beunckens, Caroline, Sotto, Cristina and Kenward, Michael G. (2008). Every Missingness Not at Random Model Has a Missingness at Random Counterpart with Equal Fit. *Journal of the Royal Statistical Society, Series B* 70(2), 371–388.
- National Research Council (2010). *The Prevention and Treatment of Missing Data in*

Clinical Trials. Washington, DC: National Academies Press.

- Orchard, Terence and Woodbury, Max A. (1972). A Missing Information Principle: Theory and Applications. *Proceedings of the Sixth Berkeley Symposium on Mathematical Statistics and Probability* 1, 697–715.
- Raghunathan, Trivellore E., Lepkowski, James M., Van Hoewyk, John and Solenberger, Peter (2001). A Multivariate Technique for Multiply Imputing Missing Values Using a Sequence of Regression Models. *Survey Methodology* 27(1), 85–95.
- Robins, James M., Rotnitzky, Andrea and Zhao, Lue Ping (1994). Estimation of Regression Coefficients When Some Regressors Are Not Always Observed. *Journal of the American Statistical Association* 89(427), 846–866.
- Rubin, Donald B. (1976). Inference and Missing Data. *Biometrika* 63(3), 581–592.
- Rubin, Donald B. (1978). Multiple Imputations in Sample Surveys — A Phenomenological Bayesian Approach to Nonresponse. *Proceedings of the Survey Research Methods Section, American Statistical Association*, 20–34.
- Rubin, Donald B. (1987). *Multiple Imputation for Nonresponse in Surveys*. New York: Wiley.
- Schafer, Joseph L. (1997). *Analysis of Incomplete Multivariate Data*. London: Chapman and Hall.
- Seaman, Shaun R., Galati, John, Jackson, Dan and Carlin, John (2013). What Is Meant by “Missing at Random”? *Statistical Science* 28(2), 257–268.
- Van Buuren, Stef (2018). *Flexible Imputation of Missing Data*. 2nd edition. Boca Raton, FL: Chapman and Hall/CRC.
- Van Buuren, Stef, Boshuizen, Hendriek C. and Knook, Dick L. (1999). Multiple Imputation of Missing Blood Pressure Covariates in Survival Analysis. *Statistics in Medicine* 18(6), 681–694.
- White, Ian R. and Carlin, John B. (2010). Bias and Efficiency of Multiple Imputation Compared with Complete-Case Analysis for Missing Covariate Values. *Statistics in Medicine* 29(28), 2920–2931.
- Wu, C. F. Jeff (1983). On the Convergence Properties of the EM Algorithm. *The Annals of Statistics* 11(1), 95–103.
- Yates, F. (1933). The Analysis of Replicated Experiments When the Field Results Are Incomplete. *Empire Journal of Experimental Agriculture* 1(2), 129–142.